

Multi-Source Candidate Data Transformer — Technical Design

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PROBLEM

Eightfold ingests candidate data from many disconnected sources — recruiter CSVs, ATS exports, resumes, free-text notes, public GitHub profiles. Downstream products need **one canonical profile per person**: fixed schema, normalized values, deduplicated arrays, field-level provenance, and honest confidence. The core design tension: a wrong value stated confidently is more damaging than a missing one, because it silently corrupts hiring decisions downstream. Every design choice below optimizes for that asymmetry — prefer null over guess, prefer explainable rules over opaque ML.

PIPELINE

Twelve explicit stages, each independently testable and replaceable:

Detect → Parse → Extract	CLI flags select inputs; per-source adapters map foreign field names (e.g. ATS <i>candidateName</i> , <i>workHistory</i>) into one common source-profile shape.
Normalize	Source-local only — each profile is normalized in isolation before any cross-source comparison happens, so merge logic never has to reconcile mixed formats.
Identity Resolution	Groups profiles believed to be the same person, using conservative deterministic keys (see right column).
Merge	Collapses a group into one canonical record: resolves scalar conflicts, unions arrays, records provenance per field.
Confidence	Per-field score from source reliability + cross-source agreement – conflict penalty.
Projection	Optional runtime JSON config reshapes the output for a given consumer without mutating the canonical record.
Validation	Three checkpoints — config (pre-run), canonical model (post-merge), projected output (pre-write) — so a bad config or bad merge never reaches a caller.

MERGE & CONFLICT-RESOLUTION POLICY

Identity keys (conservative, deterministic): email first — *email:{normalized_email}* — else *name_phone:{normalized_name}{E.164}* when both are present. Name alone is never used as a key; it collides too easily across a large candidate pool. GitHub is excluded from identity entirely — handles and public commit emails are not reliable enough to anchor a person's record; GitHub may only enrich a profile after another source has already established identity.

Scalar conflicts: fixed source-priority ladder — resume → ATS JSON → recruiter CSV → recruiter notes → GitHub. Highest-priority present value wins; the losing value is not discarded but logged into *_issues* and the field's confidence is penalized, so a human can audit the disagreement later.

Array conflicts: emails, phones, and skills are unioned in priority order rather than overwritten. Experience entries dedupe on normalized *company* + *title*; education on *institution*; once two rows are judged the same record, a lower-priority source can still fill a subfield the winner left blank (e.g. ATS *summary* merged into a resume row). Phones sharing a national number collapse into one E.164 value, with country chosen from source location context rather than guessed.

Confidence formula: base source reliability (resume 1.0, ATS 0.85, CSV 0.70, notes 0.50, GitHub 0.40) plus a bonus when independent sources agree, minus a flat 0.25 penalty per recorded conflict. Deliberately a transparent, hand-tunable heuristic rather than a learned model — every score is explainable in one sentence to a non-engineer reviewing a hiring decision.

CANONICAL SCHEMA & NORMALIZED FORMATS

Field	Normalization rule
<i>candidate_id</i>	First 16 hex chars of sha256(identity_key) — stable, source-independent ID.
<i>full_name</i>	Trimmed text only; no casing changes that could mangle names.
<i>emails[]</i>	Lower-cased.
<i>phones[]</i>	E.164 via phonenumbers lib; dedup by national number at merge time.
<i>location</i>	{city, region, country}; country mapped to ISO-3166 alpha-2.
<i>links</i>	{linkedin, github, portfolio, others} forced to HTTPS URLs.
<i>headline</i>	Trimmed text or null.
<i>years_experience</i>	Computed from non-overlapping experience-entry months; an open-ended "end" uses the pipeline run date as reference.
<i>skills[]</i>	{name, confidence, sources}; lower-cased with a small alias map (ReactJS → react, PostgreSQL → postgres).
<i>experience[]</i>	{company, title, start, end, summary}; dates normalized to YYYY-MM.
<i>education[]</i>	{institution, degree, field, end_year}; end_year coerced to integer when present.
<i>provenance[]</i>	{field, source, method} — one entry per contributing value, the audit trail for every fact.
<i>overall_confidence</i>	Mean of all populated field-level confidences.

Unknown values stay *null* / *[]* — the schema has no field that is ever fabricated to avoid a gap.

RUNTIME PROJECTION & VALIDATION

A separate, read-only layer driven by a JSON config (*configs/custom_projection.json*) reshapes the canonical record per-consumer without ever mutating it — the same canonical record can feed a search index, an ATS sync job, and a recruiter dashboard with three different shapes. The config can: select a field subset; remap paths (e.g. *emails[0]* → *primary_email*); apply per-field output normalization (E.164, canonical skill names); toggle *include_confidence* / *include_provenance*; and choose an *on_missing* policy of *null*, *omit*, or *error* per field. Validation runs three times — on the config before the run starts, on the canonical model right after merge, and on the projected output right before write — so a malformed config or a corrupted merge is caught at its own layer instead of surfacing as a confusing downstream error.

EDGE CASES / OUT OF SCOPE

Malformed CSV	Missing columns become an <i>_issues</i> entry, not a crash; the run continues if ATS or resume can still establish identity.
Bad / missing resume PDF	Extraction failure is logged as an issue; ATS JSON still supplies structured work history so the candidate isn't dropped.
Conflicting scalars	E.g. two different headlines — priority ladder picks the winner, the conflict is recorded and confidence is penalized.
Duplicate phone formats	Bare 9398172938 vs +91 9398172938 collapse to one E.164 number using location as the country hint.
GitHub-only input	Rejected as an identity anchor; GitHub can only enrich skills after a claimed source already established the person.

Deliberately deferred under time pressure — each is a real future need, not an oversight: live LinkedIn scraping (no reliable public access); ML / fuzzy name matching or skill embeddings (would trade explainability for marginal recall); arbitrary resume-layout parsing beyond conservative pdfplumber heuristics (ATS already carries structured work history); distributed batch orchestration for thousands of GitHub calls; DOCX resume support.